BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: KT-X-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code A

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. When the value of "Kc" is very small it shows that .....

(A) reaction is at equilibrium
(B) equilibrium will never be achieved
(C) reaction will move in forward direction
(D) reaction will move in backward direction

Qii. Formic acid is found in .....

(A) apples
(B) spanic
(C) grapes
(D) insects

Qiii. Which one of the following is weak acid ?.

(A) H_2SO_4
(B) HCl
(C) H_2CO_3
(D) HNO_3

Qiv. Which one of the following is aromatic compound?.

(A) anthracene
(B) furane
(C) thiophene
(D) pyridine

Qv. The natural gas is mainly composed of ....

(A) CH_4
(B) C_2H_6
(C) C_2H_4
(D) C_2H_2

Qvi. Which of the following is the least reactive?.

(A) ethyne
(B) ethane
(C) propene
(D) ethene

Qvii. On hydrolysis sucrose produces glucose with.....

(A) lactose (B) maltose

(C) fructose (D) galactose

Qviii. Anaemia causes by the deficiency of .....

(A) vitamin A (B) vitamin B

(C) vitamin C (D) vitamin E

Qix. Lowest temperature in troposphere is .....

(A) 5^0C (B) -5^0C

(C) 56^0C (D) -56^0C

Qx. Water gas is a mixture of .....

(A) CO and H_2 (B) C and H_2

(C) CO and H_2 O (D) C and H_2 O

Qxi. Sodium bicarbonate (Na HCO_3) was prepared by .....

(A) Ernest Solvay (B) Henry Cavindish

(C) Wholer (D) Newton

Qxii. Blister copper is refined through .....

(A) concentration (B) reduction

(C) smelting (D) electrolysis

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i What information are required to predict the direction of a chemical reaction?.

ii Calculate the pH and pOH of 0.001M solution of nitric acid(HNO_3).

iii Differentiate between Ali-cyclic and Aromatic organic compounds with examples.

iv Organic compound are volatile in nature why?.

v Write any four physical properties of Alkynes.

vi Write the correct equations for the preparation of Ethene from Ethyl alcohol and Ethyl chloride.

vii Classify Lipids. Give the structural formula of triglycerides and fatty acids.

viii Mention any four functions of DNA.

ix Ozone layer is depleted in the atmosphere. Explain this statement.

x Explain dielectric constant with proper mathematical formula.

xi Explain the two raw materials used for the manufacture of Urea.

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a [4] What are salts?. Give the three types of salts with proper chemical reactions.

b [3] How would you identify that dynamic equilibrium is established.

QQ.4.

a [4] What are functional groups. Support your answer with explanation of alcoholic functional group.

b [3] Compare the reactivity of alkane and alkenes.

QQ.5.

a [3] Discuss any three sources of carbohydrates.

b [4] What is air pollution ?. Describe any three man made air pollutants.

QQ.6.

a [4] How permanent hardness of water can be removed by Ion-exchange method.

b [3] Explain the origin of petroleum along with inorganic and organic origins.